

The Canonical Quantum Model of Portfolio Optimization:

A Pedagogical Introduction for Financial Practitioners

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Abstract

Portfolio optimization is one of the most natural applications of quantum computing in finance because many realistic investment problems are combinatorial, constrained, and difficult to solve at scale. This paper develops a canonical quantum formulation of portfolio optimization using the Quantum Approximate Optimization Algorithm, or QAOA. The exposition is designed for readers with advanced training in finance, economics, or quantitative methods but without a formal background in quantum physics. The central idea is to construct a mapping between a classical portfolio-selection problem and the spectral structure of a quantum Hamiltonian. Binary asset-selection variables are represented by qubits, the classical risk-return objective is converted into a quadratic unconstrained binary optimization problem, and the resulting cost function is encoded as a diagonal Hamiltonian. Quantum evolution under this Hamiltonian transforms portfolio costs into phases. A mixer Hamiltonian then allows interference to reshape the probability distribution over candidate portfolios. The paper explains each stage of this construction, including the financial objective, the binary encoding, the Pauli operators, the cost unitary, the mixer, the hybrid classical-quantum optimization loop, measurement, benchmarking, and governance. The paper is aligned with an accompanying fifteen-asset notebook that implements a depth-one statevector QAOA simulation, draws simulated measurement samples, compares the sampled portfolio with exhaustive enumeration, and records an audit package. The paper emphasizes that the current value of the framework is primarily conceptual, experimental, and architectural rather than evidence of established quantum advantage.

Notebook: [Interactive Google Colab notebook](#). A local copy is included with the project files. This paper follows the notebook’s pedagogical workflow and treats the notebook as the executable companion to the exposition.

1 Introduction

Portfolio optimization begins with a familiar economic problem: how should an investor allocate scarce capital across a set of assets when expected returns, risks, correlations, and investment constraints must be considered simultaneously? In its classical form, the problem is closely associated with the mean-variance framework introduced by [Markowitz \(1952\)](#). The investor selects a vector of portfolio weights to maximize expected return for a given level of risk, minimize risk for a required level of return, or optimize a utility function combining both objectives.

In the simplest continuous formulation, let

$$w = \begin{pmatrix} w_1 & w_2 & \cdots & w_n \end{pmatrix}^\top$$

denote the vector of portfolio weights. Let

$$\mu = \begin{pmatrix} \mu_1 & \mu_2 & \cdots & \mu_n \end{pmatrix}^\top$$

denote expected returns, and let

$$\Sigma$$

denote the covariance matrix of asset returns. Expected portfolio return is

$$\mu^\top w,$$

while portfolio variance is

$$w^\top \Sigma w.$$

A standard mean-variance objective is

$$\min_w [\lambda w^\top \Sigma w - \mu^\top w],$$

subject to constraints such as

$$\sum_{i=1}^n w_i = 1, \quad w_i \geq 0.$$

The scalar $\lambda > 0$ represents risk aversion. A larger value gives greater weight to variance reduction, while a smaller value places greater emphasis on expected return.

This continuous problem is often computationally manageable. However, real portfolio

construction frequently introduces discrete restrictions. An asset may be either included or excluded. The portfolio may contain no more than a specified number of positions. Minimum transaction sizes may be imposed. Sector exposures, turnover, regulatory limits, liquidity constraints, and trading costs may create nonlinear and combinatorial structures. In such settings, portfolio optimization may become a mixed-integer or quadratic binary problem.

Quantum optimization is especially relevant to this discrete version of the problem. The canonical quantum model does not begin by assigning a continuous portfolio weight to the amplitude of a qubit. Instead, it first converts the investment decision into binary variables. Those variables are mapped to computational-basis states, the portfolio objective is mapped to a Hamiltonian, and the Hamiltonian is used to generate a quantum circuit.

The purpose of this paper is to explain that translation carefully. The discussion proceeds from the financial problem to the quantum model without assuming prior knowledge of quantum mechanics. The emphasis is not on quantum hardware engineering, but on the mathematical architecture that makes a classical optimization problem compatible with quantum operations.

The companion notebook makes the construction concrete. It uses fifteen synthetic assets, requires exactly five selections, builds a complete cost vector over all 2^{15} candidate bit strings, simulates one layer of QAOA with a statevector, samples the final probability distribution, and compares the selected valid portfolio with the exact classical optimum. The paper therefore prioritizes pedagogy: each mathematical object is introduced first as a financial object, then as a binary optimization object, then as a quantum object, and finally as a line of computation in the notebook.

The notebook is organized as an auditable sequence of cells. Its workflow is:

Cell	Pedagogical role
1	Install and import the required packages, then print runtime versions and a UTC timestamp.
2	Define the configuration: 15 assets, 5 required selections, QAOA depth $p = 1$, risk weight 0.70, return weight 1.00, penalty 4.00, 5,000 shots, random seed 20260616, and L-BFGS-B optimization.
3	Generate reproducible synthetic expected returns and a positive-semidefinite covariance matrix from a factor structure.
4	Enumerate all bit strings, compute the QUBO cost for each one, identify the valid five-asset states, and compute the exact classical optimum.
5	Define the ideal statevector QAOA operations: a diagonal cost phase and a Pauli- X mixer.
6	Search over (γ, β) with a coarse grid and refine the parameters with L-BFGS-B while logging the trace.
7	Sample the optimized probability distribution and select the lowest-cost valid portfolio observed.
8–11	Compute financial metrics, compare against exhaustive enumeration, visualize high-probability valid portfolios, build an audit package, and generate a controlled language-model explanation.

This roadmap also clarifies the division of labor. The notebook’s numerical cells determine the portfolio; the final language-model cell only explains already-computed results and is recorded as a reporting artifact.

2 From Continuous Allocation to Binary Portfolio Selection

The most transparent quantum formulation begins with portfolio selection rather than unrestricted continuous allocation. Suppose an investment universe contains n assets. Define

$$x_i = \begin{cases} 1, & \text{if asset } i \text{ is included,} \\ 0, & \text{otherwise.} \end{cases}$$

The decision vector is

$$x = \begin{pmatrix} x_1 & x_2 & \cdots & x_n \end{pmatrix}^\top, \quad x_i \in \{0, 1\}.$$

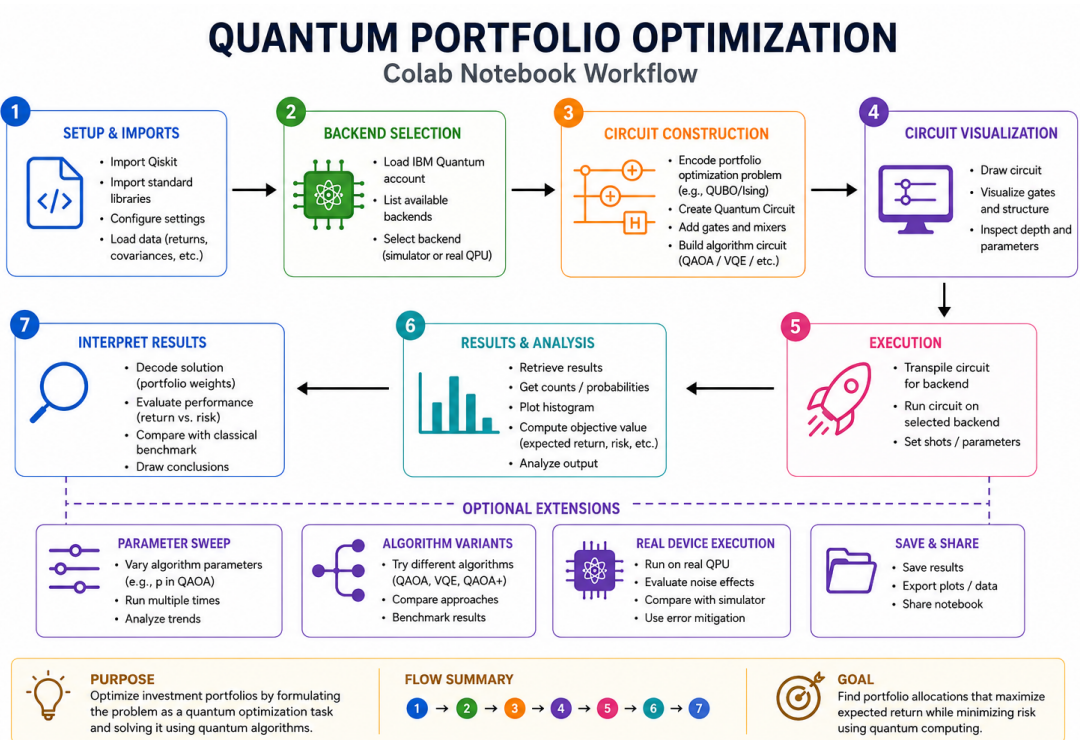


Figure 1: Pedagogical overview of the quantum portfolio-optimization workflow. The figure summarizes the movement from asset data and binary selection to QUBO construction, QAOA evolution, measurement, classical benchmarking, and auditability.

The mathematical notation in this paper numbers assets from 1 to n . The companion notebook uses Python arrays, so the same assets are stored at indices 0 to 14. This is only an indexing convention. A one still means “include the asset,” and a zero still means “exclude the asset.”

If exactly K assets must be selected, then

$$\sum_{i=1}^n x_i = K.$$

A simple allocation rule is to assign equal weights to the selected positions:

$$w_i = \frac{x_i}{K}.$$

Under this convention, the expected portfolio return is

$$R(x) = \frac{1}{K} \mu^\top x,$$

and variance is

$$V(x) = \frac{1}{K^2} x^\top \Sigma x.$$

Since the constants $1/K$ and $1/K^2$ can be absorbed into model coefficients, a convenient binary objective is

$$C(x) = \lambda x^\top \Sigma x - \eta \mu^\top x,$$

where λ controls the risk penalty and η controls the return reward.

The exact- K constraint can be included through a quadratic penalty:

$$P \left(\sum_{i=1}^n x_i - K \right)^2,$$

where $P > 0$ is chosen sufficiently large to make invalid portfolios unattractive. The complete objective becomes

$$C(x) = \lambda x^\top \Sigma x - \eta \mu^\top x + P \left(\sum_{i=1}^n x_i - K \right)^2.$$

This expression is a quadratic unconstrained binary optimization problem, usually abbreviated as QUBO. It is “unconstrained” only in the technical sense that the constraint has been absorbed into the objective as a penalty. The economic restriction remains present.

In the notebook, this formula is implemented directly as a score over selected assets:

$$C(x) = 0.70 x^\top \Sigma x - 1.00 \mu^\top x + 4.00 \left(\sum_{i=1}^{15} x_i - 5 \right)^2.$$

The constants 0.70, 1.00, and 4.00 are the notebook’s risk-aversion, return-reward, and cardinality-penalty coefficients. After a bit string is selected, the notebook separately converts it into an equal-weight portfolio by assigning weight 1/5 to each selected asset. Thus, the QUBO is the selection score, while the reported return, volatility, and Sharpe ratio are conventional financial metrics computed after selection.

The portfolio problem is now fully described by a cost assigned to each binary string. With three assets, the string

$$x = (1, 0, 1)$$

means that assets 1 and 3 are selected. In quantum notation, the same portfolio is written

$$|101\rangle.$$

With fifteen assets, the model uses fifteen binary variables and therefore fifteen qubits. The full search space contains

$$2^{15} = 32,768$$

possible bit strings. If exactly five assets must be selected, the number of valid portfolios is

$$\binom{15}{5} = 3,003.$$

This size remains small enough for exhaustive classical benchmarking, making it ideal for pedagogical experimentation.

3 The Quantum Encoding

The quantum encoding starts with one qubit per binary decision. A qubit has two basis states used for computation:

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

For portfolio selection, these states are interpreted as

$$|0\rangle \longleftrightarrow \text{asset excluded},$$

$$|1\rangle \longleftrightarrow \text{asset included}.$$

For n assets, the full quantum register is

$$|x_1 x_2 \cdots x_n\rangle.$$

The computational basis therefore provides a direct representation of all classical portfolios.

The next challenge is to represent the binary variable x_i as a quantum operator. The relevant operator is the Pauli- Z matrix:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Its action on the computational basis is

$$Z|0\rangle = |0\rangle,$$

$$Z|1\rangle = -|1\rangle.$$

From this operator, define

$$\hat{x}_i = \frac{I - Z_i}{2},$$

where I is the identity operator and Z_i acts on qubit i .

The operator has the desired binary behavior:

$$\hat{x}_i|0\rangle = 0|0\rangle,$$

$$\hat{x}_i|1\rangle = 1|1\rangle.$$

Thus, $\hat{x}_i$ is the operator representation of the classical variable x_i .

For the three-asset portfolio state

$$|101\rangle,$$

the operators satisfy

$$\hat{x}_1|101\rangle = |101\rangle,$$

$$\hat{x}_2|101\rangle = 0,$$

$$\hat{x}_3|101\rangle = |101\rangle.$$

The eigenvalues are therefore

$$(1, 0, 1),$$

which reproduce the original classical portfolio.

This is the essential mapping:

$$x_i \longrightarrow \frac{I - Z_i}{2}.$$

Linear terms in the portfolio objective become single-qubit Z_i terms. Quadratic terms such as $x_i x_j$, which arise from covariance and penalties, become combinations containing $Z_i Z_j$ interactions.

4 The Cost Hamiltonian

After replacing every binary variable with its operator equivalent, the classical objective becomes a quantum operator:

$$H_C = C\left(\frac{I - Z_1}{2}, \frac{I - Z_2}{2}, \dots, \frac{I - Z_n}{2}\right).$$

This operator is called the cost Hamiltonian.

The terminology may initially appear unusual to a financial reader. In physics, a Hamiltonian is the operator associated with the energy of a system. In optimization, the concept is used more abstractly. The Hamiltonian is engineered so that the energy of each computational-basis state equals the cost of the corresponding portfolio:

$$H_C|x\rangle = C(x)|x\rangle.$$

If a particular portfolio has cost -15 , then

$$H_C|x\rangle = -15|x\rangle.$$

If another portfolio has cost -12 , then

$$H_C|y\rangle = -12|y\rangle.$$

The portfolios themselves are unchanged. The operator associates each state with its classical objective value.

This construction creates an embedding of the classical optimization landscape into the spectrum of a quantum operator. Low-cost portfolios correspond to low-energy eigenstates. The optimization objective can therefore be expressed as the search for a low-energy state of H_C .

The mapping is designed by the modeler, but it is not arbitrary. The cost Hamiltonian is required to reproduce the original objective function on every computational-basis state. If it does not, the quantum model is solving a different problem.

The distinction between the financial problem and the quantum representation should be maintained carefully. The expected returns and covariance matrix originate in finance. The QUBO is a binary reformulation. The Hamiltonian is the quantum operator representation of that QUBO. The economic content is not created by the quantum computer; it is supplied through the model.

For pedagogy, the notebook uses the most direct statevector representation of this idea. Instead of first expanding the QUBO into an explicit sum of Pauli strings, it enumerates every state identifier from 0 to $2^{15} - 1$, decodes each identifier into a binary portfolio vector, computes $C(x)$, and stores the resulting diagonal vector. In mathematical form, the simulated Hamiltonian is

$$H_C = \sum_{x \in \{0,1\}^{15}} C(x)|x\rangle\langle x|.$$

This diagonal form is equivalent, for the purpose of an ideal statevector simulation, to a cost Hamiltonian whose computational-basis eigenvalues are the portfolio costs. A hardware implementation would usually compile the QUBO into one- and two-qubit Pauli- Z interactions, but the notebook keeps the diagonal representation visible so that the finance-to-quantum mapping can be inspected directly.

5 From Hamiltonian to Quantum Evolution

A Hamiltonian is not generally a valid quantum gate because it need not preserve probability. Quantum transformations must be unitary. Given a Hamiltonian H , quantum mechanics associates it with the unitary evolution

$$U(t) = e^{-itH},$$

where units are chosen so that $\hbar = 1$.

In QAOA, the cost unitary is

$$U_C(\gamma) = e^{-i\gamma H_C}.$$

The parameter γ plays a role analogous to an evolution time. Since each portfolio state is an eigenstate of the cost Hamiltonian,

$$H_C|x\rangle = C(x)|x\rangle,$$

it follows that

$$U_C(\gamma)|x\rangle = e^{-i\gamma C(x)}|x\rangle.$$

This equation is central to the entire algorithm. The portfolio remains the same bit string, but its amplitude acquires a phase determined by its cost.

In the notebook this operation appears as an elementwise multiplication of the statevector:

$$a_x \longrightarrow a_x e^{-i\gamma C(x)}.$$

This line is often the easiest way to understand the cost unitary. The simulator already knows the cost of every portfolio, so it multiplies each portfolio amplitude by the corresponding complex phase.

For example, suppose

$$C(101) = -15.$$

Then

$$U_C(\gamma)|101\rangle = e^{i15\gamma}|101\rangle.$$

Another portfolio with cost -12 receives the phase

$$e^{i12\gamma}.$$

At this stage, the probabilities have not changed. Multiplying an amplitude by a complex number of magnitude one changes only its phase:

$$|e^{-i\gamma C(x)} a_x|^2 = |a_x|^2.$$

Therefore, cost encoding alone cannot make a good portfolio more likely to be measured. A second transformation is required.

6 The Mixer Hamiltonian

The standard QAOA mixer Hamiltonian is

$$H_M = \sum_{i=1}^n X_i,$$

where

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is the Pauli- X operator. It exchanges the computational-basis states:

$$X|0\rangle = |1\rangle,$$

$$X|1\rangle = |0\rangle.$$

The corresponding mixer unitary is

$$U_M(\beta) = e^{-i\beta H_M}.$$

The mixer allows probability amplitude to move among different binary portfolios. Because the cost unitary has assigned different phases to different states, the mixer causes those amplitudes to interfere.

For one qubit, the mixer unitary has the form

$$e^{-i\beta X} = \begin{pmatrix} \cos \beta & -i \sin \beta \\ -i \sin \beta & \cos \beta \end{pmatrix}.$$

Thus, for a pair of amplitudes that differ only in one bit,

$$\begin{pmatrix} a'_0 \\ a'_1 \end{pmatrix} = \begin{pmatrix} \cos \beta & -i \sin \beta \\ -i \sin \beta & \cos \beta \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}.$$

The notebook implements this operation by sweeping through the fifteen qubits and applying this two-amplitude update to the appropriate pairs in the statevector.

Constructive interference can increase the probability of some portfolios. Destructive interference can reduce the probability of others. Optimization arises from the combined sequence:

$$\text{cost-dependent phase} \longrightarrow \text{mixing} \longrightarrow \text{interference}.$$

The simplest QAOA state, with depth $p = 1$, is

$$|\psi(\gamma, \beta)\rangle = U_M(\beta)U_C(\gamma)|+\rangle^{\otimes n},$$

where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

The initial state is

$$|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle.$$

Every possible portfolio begins with equal amplitude.

For depth $p > 1$, the cost and mixer transformations are repeated:

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{\ell=1}^p U_M(\beta_\ell)U_C(\gamma_\ell)|+\rangle^{\otimes n}.$$

Greater depth increases expressive capacity but also increases circuit length, parameter count, optimization difficulty, and exposure to hardware noise.

7 The Hybrid Optimization Loop

The values of γ and β are not known in advance. They are chosen by a classical optimizer.

For a trial parameter vector, the quantum circuit prepares

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle.$$

The expected portfolio cost is

$$F(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) | H_C | \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) \rangle.$$

The classical optimizer seeks

$$\min_{\boldsymbol{\gamma}, \boldsymbol{\beta}} F(\boldsymbol{\gamma}, \boldsymbol{\beta}).$$

The process is iterative:

1. Select trial QAOA parameters.
2. Prepare the quantum state.
3. Apply the cost and mixer unitaries.
4. Estimate the expected cost.
5. Update the parameters using a classical optimizer.
6. Repeat until a stopping rule is reached.

This structure explains why QAOA is called a hybrid quantum-classical algorithm. The quantum circuit constructs and evaluates a parameterized probability distribution. The classical optimizer governs the search over circuit parameters.

For a statevector simulation, the expectation can be calculated exactly:

$$F(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \sum_x p_{\boldsymbol{\gamma}, \boldsymbol{\beta}}(x) C(x),$$

where

$$p_{\boldsymbol{\gamma}, \boldsymbol{\beta}}(x) = |\langle x | \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) \rangle|^2.$$

On actual hardware, the distribution must be estimated from repeated circuit executions, usually called shots.

The companion notebook uses the statevector version of this loop. For each trial $(\boldsymbol{\gamma}, \boldsymbol{\beta})$, it constructs the full vector of 32,768 amplitudes, squares their magnitudes to obtain exact ideal probabilities, and evaluates

$$\sum_x p_{\gamma,\beta}(x)C(x)$$

without sampling noise. It first searches a 13×13 grid over $\gamma \in [0, 2\pi]$ and $\beta \in [0, \pi]$, then refines the best grid point with L-BFGS-B. The optimized parameters recorded in the notebook are approximately

$$\gamma = 6.283185307, \quad \beta = 1.032047006.$$

The optimizer's expected cost is the expectation over the full ideal distribution, including invalid cardinalities if they still have nonzero probability. This quantity is not the same as the cost of the final selected five-asset portfolio.

8 Measurement and Portfolio Selection

After the QAOA parameters have been optimized, the circuit is executed repeatedly. Each measurement returns one classical bit string:

$$x^{(1)}, x^{(2)}, \dots, x^{(S)}.$$

Each bit string is an ordinary portfolio-selection vector.

For example,

$$|101000100010001\rangle$$

indicates that the assets corresponding to the positions containing ones have been selected.

The output of the algorithm is therefore not a quantum portfolio. It is a probability distribution over classical portfolios. A final selection rule must be specified. Possible choices include:

- the most frequently measured valid portfolio;
- the lowest-cost valid portfolio observed;
- the best portfolio among the top- m measured states;
- a portfolio selected after additional classical refinement.

The accompanying notebook uses the lowest-cost valid portfolio observed in the simulated sample. This distinction matters because the most probable portfolio is not necessarily the lowest-cost one.

The notebook uses $S = 5,000$ simulated shots. It samples state identifiers from the optimized statevector probabilities, discards states that do not contain exactly five selected assets when choosing the final portfolio, and then selects the observed valid state with the lowest cost. In the recorded run, the selected QAOA bit vector was

$$(0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0),$$

corresponding to

$$\{\text{Asset_03}, \text{Asset_05}, \text{Asset_07}, \text{Asset_10}, \text{Asset_13}\}.$$

Its QUBO cost was

$$C(x_{\text{QAOA}}) = -0.5159032991.$$

This state appeared once in the 5,000 simulated shots. That count should not be overinterpreted as a production confidence measure; it is a pedagogical sample from an ideal statevector distribution.

Once the binary portfolio is selected, standard financial quantities can be calculated. Under equal weighting,

$$w_i = \frac{x_i}{K}.$$

Expected return is

$$\mu^\top w,$$

variance is

$$w^\top \Sigma w,$$

and volatility is

$$\sqrt{w^\top \Sigma w}.$$

A simplified Sharpe ratio can be calculated as

$$\frac{\mu^\top w - r_f}{\sqrt{w^\top \Sigma w}}.$$

Thus, quantum measurement returns the portfolio structure, while conventional financial

analysis evaluates its economic characteristics.

9 The Fifteen-Asset Canonical Example

The companion notebook implements a fifteen-asset model. The inputs consist of a synthetic annual expected-return vector and a positive-semidefinite covariance matrix generated from a three-factor structure with idiosyncratic variances. A fixed random seed, 20260616, ensures that the same returns, volatilities, covariance matrix, optimization trace, and simulated samples can be reproduced.

The model selects exactly five assets. Its configuration is:

Quantity	Notebook value
Number of assets	15
Required selected assets	5
QAOA depth	$p = 1$
Risk-aversion coefficient	0.70
Return-reward coefficient	1.00
Cardinality penalty	4.00
Simulated shots	5,000
Optimizer	L-BFGS-B after a deterministic coarse grid
Random seed	20260616

The objective is

$$C(x) = 0.70 x^\top \Sigma x - 1.00 \mu^\top x + 4.00 \left(\sum_{i=1}^{15} x_i - 5 \right)^2.$$

The program explicitly constructs all

$$2^{15} = 32,768$$

portfolio states and calculates their costs. The complete cost vector acts as the diagonal representation of H_C .

The initial state is a uniform superposition over all states. The cost unitary applies

$$e^{-i\gamma C(x)}$$

to each amplitude. The mixer is implemented by sequential single-qubit X rotations. A deterministic grid search produces an initial estimate of γ and β , followed by numerical refinement using L-BFGS-B. The final recorded parameters are $\gamma \approx 2\pi$ and $\beta \approx 1.032$.

The final statevector provides an exact idealized probability distribution. A simulated measurement process draws repeated samples from that distribution. The lowest-cost valid portfolio observed is then compared with the exact classical optimum.

Because there are only 3,003 valid five-asset combinations, the exact optimum can be computed through exhaustive search. This is an important control. The QAOA result should not be evaluated without reference to a classical benchmark.

The recorded notebook run selected the following QAOA portfolio and benchmark:

Quantity	Sampled QAOA portfolio	Exact classical optimum
Assets	Asset_03, Asset_05, Asset_07, Asset_10, Asset_13	Asset_03, Asset_05, Asset_09, Asset_10, Asset_15
Bit string	001010100100100	001010001100001
Cost	−0.5159032991	−0.5637158849
Expected return	13.6633%	15.5191%
Volatility	9.7764%	11.0126%
Sharpe ratio, $r_f = 0$	1.3976	1.4092

Define the optimality gap as

$$\Delta = C(x_{\text{QAOA}}) - C(x_{\text{exact}}).$$

A value of

$$\Delta = 0$$

means that the sampled QAOA portfolio matches the exact solution. A positive value indicates a suboptimal result.

For the recorded run,

$$\Delta = -0.5159032991 - (-0.5637158849) = 0.0478125858.$$

The exact solution has a lower cost. Therefore the notebook demonstrates the workflow and the benchmarking discipline, not a superior solution method.

The notebook also creates an audit package containing the configuration, random seed, financial inputs, encoded problem, optimizer trace, sampled results, exact benchmark, software

environment, figures, and SHA-256 file hashes. A final language-model cell uses Claude Haiku to explain the structure and results. The LLM does not participate in portfolio selection and is treated strictly as a reporting layer.

10 Economic Interpretation

The quantum model does not eliminate the fundamental weaknesses of portfolio optimization. Its output remains dependent on the economic inputs and assumptions.

Expected returns are difficult to estimate and often dominate optimization error. Covariance matrices may be unstable, especially when the number of assets is large relative to the available sample. Penalty coefficients influence the effective geometry of the objective. Equal weighting among selected assets is a simplifying assumption rather than a universally optimal allocation rule.

The quantum algorithm therefore solves the encoded problem, not the true investment problem in an absolute sense. If expected returns are poorly estimated, the quantum solution may be precisely optimal for an economically misspecified objective.

This point is particularly important for financial practitioners. Quantum computing changes the computational representation and search mechanism. It does not remove model risk.

A production model would need to consider:

- robust or Bayesian expected-return estimates;
- covariance shrinkage or factor models;
- transaction costs and turnover;
- liquidity constraints;
- sector and issuer limits;
- minimum and maximum allocations;
- stress testing;
- estimation uncertainty;
- out-of-sample performance;
- sensitivity to the penalty coefficient;
- comparison with strong classical solvers.

11 Governance and Auditability

Governance is essential because QAOA combines multiple layers of modeling:

financial assumptions $\rightarrow$ binary formulation $\rightarrow$ Hamiltonian $\rightarrow$ circuit
 $\rightarrow$ classical optimizer $\rightarrow$ measurement rule.

Each layer introduces assumptions and potential failure modes.

The financial layer determines expected returns, covariance, constraints, and risk preferences. The QUBO layer determines how those quantities are scaled and penalized. The quantum layer determines circuit depth, mixer structure, and parameterization. The classical optimization layer determines initial conditions, stopping rules, and numerical robustness. The measurement layer determines how a final portfolio is extracted from a probability distribution.

A governable quantum portfolio model should record at least:

- the exact financial inputs;
- the cost-function coefficients;
- the binary encoding;
- the Hamiltonian construction;
- the QAOA depth;
- the optimized parameters;
- the optimizer trace;
- the random seeds;
- the number of shots;
- the portfolio-selection rule;
- the classical benchmark;
- the software and hardware environment;
- all changes between model versions.

The use of a language model for explanation should also be controlled. The prompt, response, model identifier, temperature, timestamp, and token usage should be stored. Most importantly, the LLM should not silently modify numerical results or introduce unsupported claims.

12 Limitations and the Question of Quantum Advantage

The fifteen-asset example does not demonstrate quantum advantage. It runs on a classical statevector simulator, and the exact optimum can be obtained rapidly by exhaustive enumeration.

Statevector simulation itself scales exponentially. An n -qubit state requires storage of 2^n complex amplitudes. At fifteen qubits this is easy. At fifty or one hundred qubits it becomes impractical on ordinary classical hardware.

Quantum hardware avoids storing the full statevector explicitly, but present devices introduce noise, gate errors, limited connectivity, finite shot counts, and short coherence times. These limitations may prevent the ideal QAOA distribution from being realized.

There is also no general result establishing that low-depth QAOA will outperform the best classical portfolio solvers. Classical mixed-integer quadratic programming, branch-and-bound methods, simulated annealing, tabu search, genetic algorithms, and problem-specific heuristics provide strong benchmarks.

The responsible interpretation is therefore that QAOA offers a new computational architecture for representing and exploring combinatorial portfolio problems. Whether that architecture eventually delivers economically meaningful speed, scale, or solution-quality advantages remains an empirical and theoretical research question.

13 Extensions

Several extensions follow naturally from the canonical model.

First, portfolio weights can be discretized rather than restricted to simple inclusion or exclusion. If asset i can receive one of several allocation levels, multiple qubits can encode its weight. For example,

$$w_i = \delta \sum_{b=0}^{m-1} 2^b x_{ib},$$

where δ is the minimum allocation increment.

Second, the standard mixer may be replaced with a constraint-preserving mixer that keeps the system within the subspace containing exactly K selected assets. This can reduce the need for large penalty terms.

Third, the QAOA depth can be increased:

$$p = 2, 3, \dots$$

allowing more expressive states. However, deeper circuits introduce more parameters and may suffer from difficult optimization landscapes.

Fourth, the quantum solution can be embedded in a hybrid workflow. QAOA may generate promising candidate portfolios, which are then refined through a classical local optimizer or subjected to additional financial constraints.

Fifth, the model may be extended to multi-period portfolio optimization, transaction-cost minimization, index tracking, credit portfolio construction, or capital budgeting.

14 Conclusion

The canonical quantum model of portfolio optimization is built through a deliberate mapping between classical financial decisions and quantum mathematical structures. A binary portfolio vector is represented by a computational-basis state. The classical risk-return objective is transformed into a QUBO. Binary variables are replaced by operators of the form

$$\frac{I - Z_i}{2}.$$

The resulting cost Hamiltonian satisfies

$$H_C|x\rangle = C(x)|x\rangle.$$

Quantum evolution under this Hamiltonian produces

$$U_C(\gamma)|x\rangle = e^{-i\gamma C(x)}|x\rangle,$$

thereby encoding portfolio costs as phases. A mixer then converts phase differences into interference and changes the measurement probabilities of candidate portfolios.

The optimization process is hybrid. A classical algorithm adjusts the QAOA parameters, while the quantum circuit generates a parameterized probability distribution over portfolios. Measurement returns ordinary classical bit strings, which are decoded into asset selections

and evaluated using conventional portfolio metrics.

The framework is valuable because it provides a coherent mathematical apparatus for translating combinatorial financial optimization into quantum computation. It also reveals that the difficult questions of portfolio construction remain present. Expected returns must still be estimated. Risk must still be modeled. Constraints must still be selected. Penalty coefficients must still be calibrated. Results must still be benchmarked.

For non-physicists, the deepest conceptual insight is that the Hamiltonian is the quantum representation of the portfolio scoring rule. The cost unitary converts those scores into phases, and the mixer uses interference to redistribute probabilities. The algorithm does not examine every portfolio and print the best one. It engineers a probability distribution in which favorable portfolios are intended to become easier to observe.

For financial researchers, the principal value of the model lies in its architecture. It establishes a disciplined chain from economic assumptions to binary optimization, from optimization to operator theory, and from operator theory to a measurable circuit. That chain can be governed, logged, benchmarked, and audited.

The current generation of quantum portfolio models should therefore be viewed as experimental research systems rather than replacements for established portfolio optimizers. Their significance lies in showing how financial optimization can be expressed in a fundamentally different computational language. Whether that language will eventually provide practical advantage depends on advances in hardware, algorithms, error correction, parameter optimization, and problem-specific design.

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